

# Exploring a Synthetic SSc Cohort

Data Understanding, Application, and Data Quality

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2026-09-11

# Semion Leyn, PhD

- Production grade bioinformatics pipelines (HPC and Cloud)
- Data solutions co-development with clinical partners (e.g. Roswell Park - NYS CLEP validation).
- Worked in:
  - Russian Academy of Sciences (academy)
  - SBP Medical discovery institute (academy)
  - Agilent Technologies (industry)

# Agenda

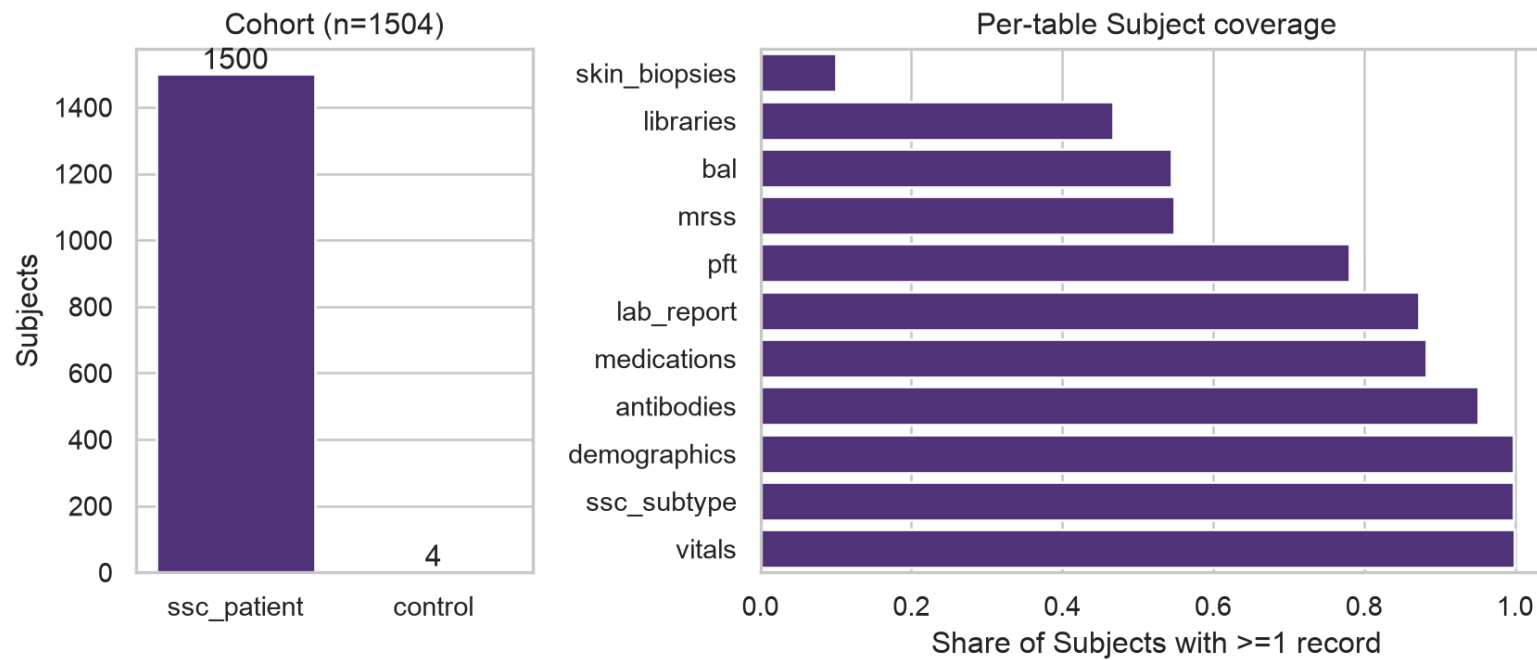
- Understanding the dataset
- The application & visualizations
- Data-quality issues and analytical findings
- Design decisions and tradeoffs
- Future work & discussion

# Understanding the Dataset

# What's in the box

- 11 CSVs, ~4 MB, synthetic and non-PHI
- A patient/subject identifier under **4 different column names** across files — normalized to one canonical `subject_id` at ingestion
- The **Registry**: `demographics` + `ssc_subtype`, a 1:1, 1,500-row master list — every row `study=SSC-REG`, `diagnosis=SSc`
- Every other table: a **partial-coverage** table of clinical/molecular events for a subset of those 1,500 patients

# Population and table coverage



# A hidden second cohort

4 subject IDs (SSC\_NORM\_0101/0102/0104/0110) appear in `libraries.csv`  
— but **never** in the Registry.

- `libraries.csv`'s own comments confirm these are “healthy control sample[s]”
- Modeled explicitly as an attribute (`ssc_patient` vs `control`)
- Not a data error
- The QC key-linkage check automates this: confirms all 4 against the known set, and would flag any *future* out-of-Registry `subject_id`.

# Static vs. longitudinal, and a naming trap

- `lab_report`, `vitals`, `mrss`, `bal`, `medications`, `skin_biopsies` — structured longitudinal event records.
- `demographics/ssc_subtype` — static, one row per patient



# The Application

# Streamlit, 4 tabs, SQLite-backed

- **Data & Dictionary** — schema, coverage, QC report link
- **Cohort Overview** — population-level distributions
- **Compare & Discover** — free-form variable picker across any table/level, with a Control Subject overlay toggle
- **Patient Trajectory** — one/several patient's longitudinal record
- **QC Report** - data QC metrics generated on the fly

Login + data encryption + audit logging gated behind a published demo account (details in Design Decisions).

[nw-ssc-de.streamlit.app](https://nw-ssc-de.streamlit.app)

# Data & Dictionary

Data & Dictionary

Cohort Overview

Compare & Discover

Patient Trajectory

QC Report

Logged in as demo

## Data & Dictionary

Every table in the built Subject/Cohort store, with live row/column counts, a plain-language description of each field informed by the QC investigation, and up to 3 real example values pulled live from the store.

|                                                                                                                         |                                                                     |                                             |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------|
| ▼ subjects — 1,504 rows x 2 columns                                                                                     |                                                                     |                                             |
| Every Subject referenced anywhere in the dataset — the supertype spanning both SSc Patients and the 4 Control Subjects. |                                                                     |                                             |
| field                                                                                                                   | description                                                         | examples                                    |
| subject_id                                                                                                              | Canonical Subject identifier, normalized at ingestion.              | SSC_NORM_0101, SSC_NORM_0102, SSC_NORM_0104 |
| cohort                                                                                                                  | 'ssc_patient' or 'control' -- which Cohort this Subject belongs to. | control, ssc_patient                        |
| › demographics — 1,500 rows x 12 columns                                                                                |                                                                     |                                             |
| › ssc_subtype — 1,500 rows x 7 columns                                                                                  |                                                                     |                                             |
| › vitals — 40,250 rows x 4 columns                                                                                      |                                                                     |                                             |
| › lab_report — 30,941 rows x 4 columns                                                                                  |                                                                     |                                             |
| › mras — 3,113 rows x 4 columns                                                                                         |                                                                     |                                             |
| › pft — 5,069 rows x 5 columns                                                                                          |                                                                     |                                             |
| › antibodies — 3,511 rows x 4 columns                                                                                   |                                                                     |                                             |
| › medications — 3,273 rows x 4 columns                                                                                  |                                                                     |                                             |
| › bal — 1,170 rows x 7 columns                                                                                          |                                                                     |                                             |
| › libraries — 918 rows x 24 columns                                                                                     |                                                                     |                                             |
| › skin_biopsies — 215 rows x 8 columns                                                                                  |                                                                     |                                             |

Data & Dictionary tab: every table’s row/column counts, plain-language field descriptions, and live example values

# Cohort Overview

Data & Dictionary

Cohort Overview

Compare & Discover

Patient Trajectory

QC Report

Logged in as demo

## Cohort Overview

Population-level structure across the Registry and its 4 Control Subjects — demographic distributions, SSc subtype breakdown, and how many Subjects have at least one record in each clinical table.

Registry patients (ssc\_patient)

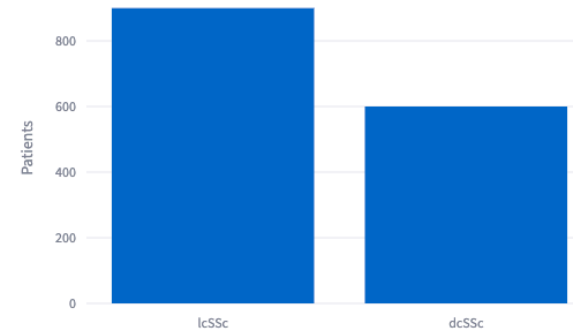
1500

Control Subjects

4

## SSc subtype

dcSSc / lcSSc split (n=1,500 Registry patients)



Cohort Overview tab

# Compare & Discover

Data & Dictionary

Cohort Overview

Compare & Discover

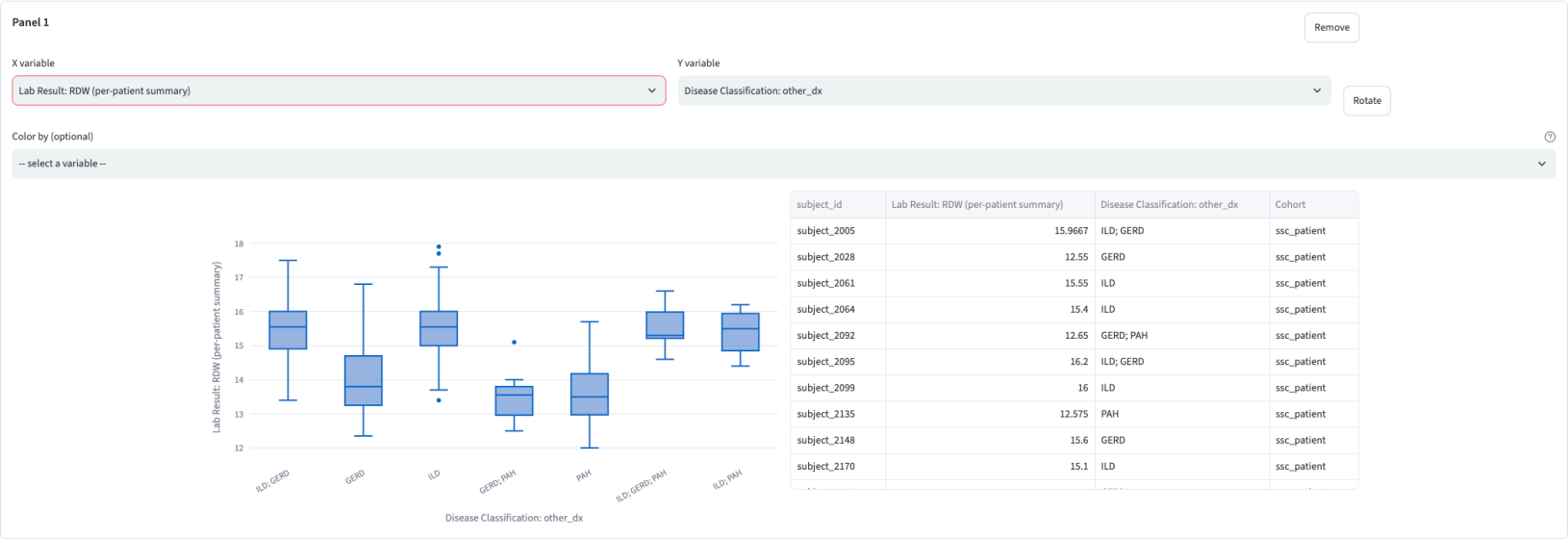
Patient Trajectory

QC Report

Logged in as demo

## Compare & Discover

Add one comparison panel per pair of variables you want to see — pick an X and a Y independently in each panel, rotate to swap them, and get the chart type that fits what you picked automatically: a scatter for two numeric measures, a box plot for a numeric measure grouped by a category, or a heatmap for two categorical fields.



Add comparison panel

Compare & Discover tab

# Patient Trajectory

Data & Dictionary

Cohort Overview

Compare & Discover

**Patient Trajectory**

QC Report

Logged in as demo

**Subject Filters**

Selected values within one filter combine with OR (e.g. picking 2 genders shows Subjects matching either); separate filters combine with AND (e.g. a gender filter plus BAL performed narrows to Subjects matching both).

> Demographics (7 filters)

**Time**

Visit date

2014-10-272018-07-13

**Medication**

Ever prescribed

Choose options

**Antibody Test**

Test result

Choose options

**BAL**

☐ BAL performed

> Lab Report (19 filters)

## Patient Trajectory

Select up to 8 Subjects by `subject_id` to see their longitudinal records overlaid — labs, vitals, MRSS, PFT, and medications plotted over time. Patient name and birth date are never shown here or anywhere else in this app.

subject\_id

subject\_2064

subject\_2092

subject\_2135

subject\_2151

subject\_2160

X-axis

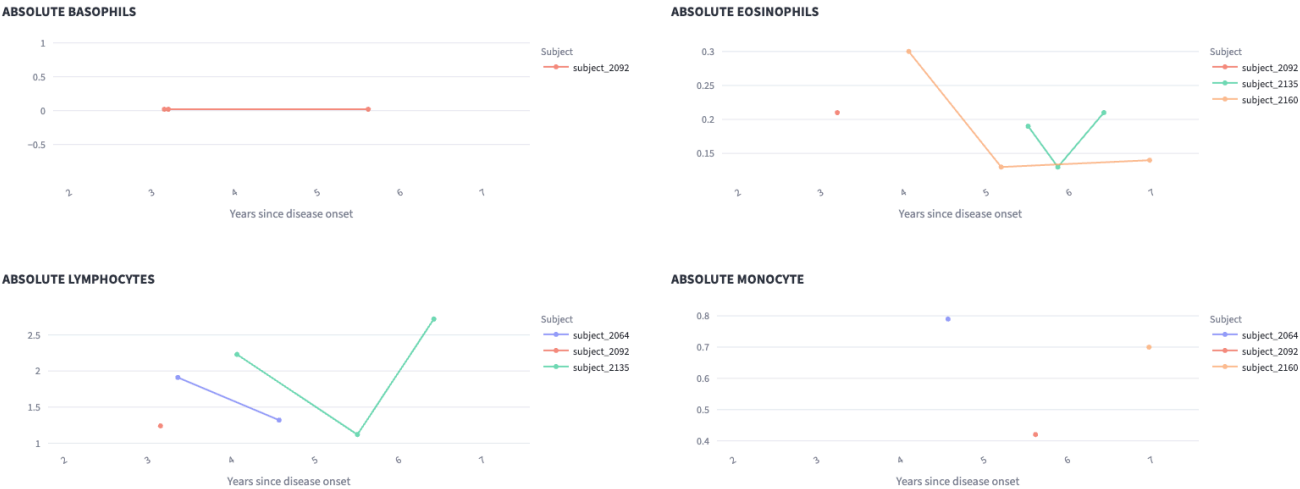
☐ Calendar date

☒ Disease duration (years since onset — first non-Raynaud symptom)

## Subject overview

| Subject      | SSc subtype | Comorbidities | Antibodies                                                                          | Disease onset | Disease duration (years) |
|--------------|-------------|---------------|-------------------------------------------------------------------------------------|---------------|--------------------------|
| subject_2064 | dcSSc       | ILD           | anti-centromere antibodies: negative; rna polymerase III: negative; scl70: negative | 2015-05-04    | 11.4                     |
| subject_2092 | lcSSc       | GERD; PAH     | anti-centromere antibodies: positive; rna polymerase III: negative                  | 2014-12-09    | 11.8                     |
| subject_2135 | lcSSc       | PAH           | anti-centromere antibodies: positive; rna polymerase III: negative                  | 2014-08-20    | 12.1                     |
| subject_2151 | dcSSc       | --            | anti-centromere antibodies: negative; scl70: positive                               | 2015-12-05    | 10.8                     |
| subject_2160 | dcSSc       | --            | scl70: positive                                                                     | 2013-12-18    | 12.7                     |

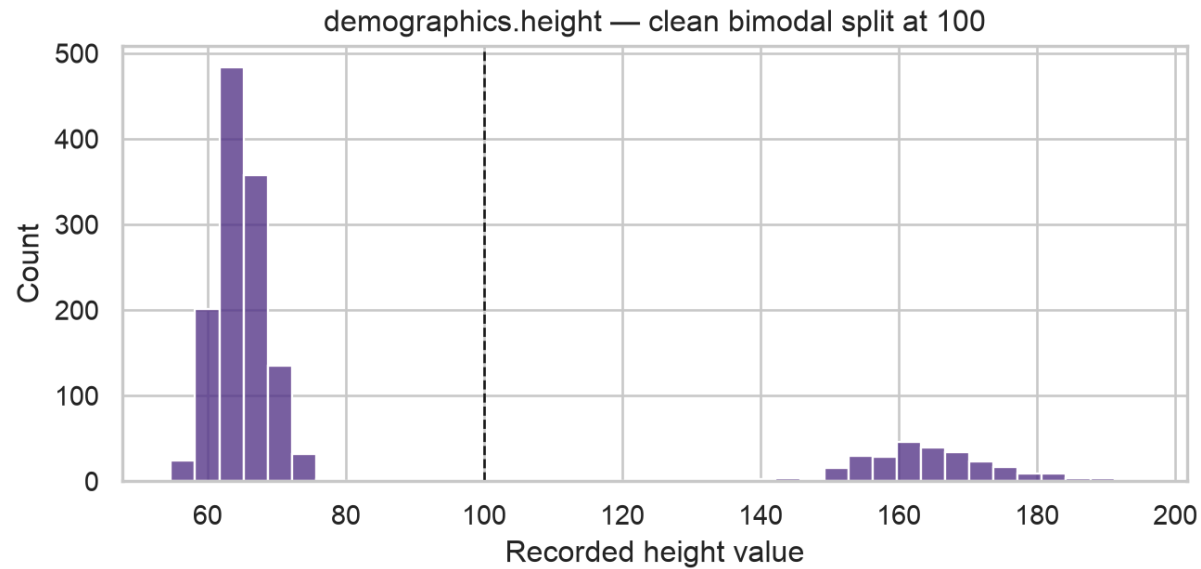
## Labs



Patient Trajectory tab

# **Data-Quality Issues and Analytical Findings**

# Height recorded in mixed units



1,235 of 1,500 patients in **inches** (54.7-78.5); the other 265 in **centimeters** (141.3-194.7 cm). Fixed at ingestion: cm-scale values converted to inches, raw CSV left untouched.

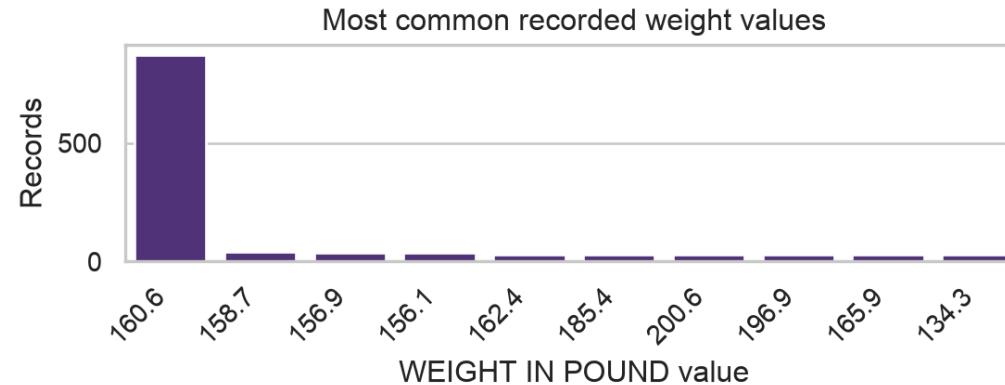


# One high-confidence data-entry error

`demographics.csv` lists `subject_8545`'s weight as **22.5** (impossible for a living adult). All 5 separate `vitals.csv` visits spanning 2017-2020 report ~160.6 lbs for the same patient - likely placeholder.

**Decision:** flagged in the QC report without correction or drop.

# A synthetic-data placeholder, not a clinical signal



`WEIGHT IN POUND = 160.6` appears 870 times across 195 distinct patients — the next most common value appears only 39 times. Some patients show this exact value on *every* visit across multiple years.

`check_constant_value_across_visits`: flags a value dominating far more patients than expected.

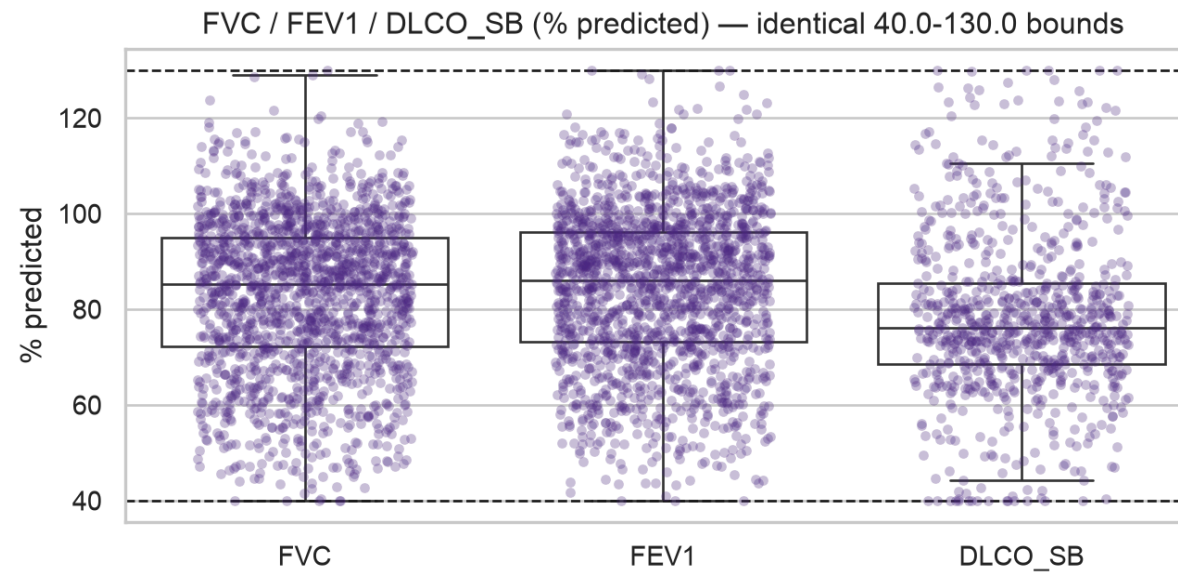
`check_value_spike`: a needle sitting on an otherwise ordinary distribution.

# The same pattern in BP and pulse, measured differently

The per-patient check above is blind to a filler used on only *some* of a patient's visits. Looking at each value's share of *every* record instead:

| Measure      | Value | Share of all records | vs. local background |
|--------------|-------|----------------------|----------------------|
| BP SYSTOLIC  | 124   | 25% of 7,848         | 11.7x                |
| BP DIASTOLIC | 77    | 25% of 7,848         | 5.7x                 |
| PULSE        | 76    | 16% of 8,381         | 4.6x                 |

# PFT bounds look like generator clipping



Three physiologically unrelated measures all range **exactly 40.0-130.0%**.

# The same generic check finds real injected errors

Running `check_guideline_range` against `lab_report.csv` surfaces isolated values:

- `ABSOLUTE LYMPHOCYTES, MCH, MCHC` — one 9999
- `WBC` — two 999
- `HEMOGLOBIN, NEUTROPHILS, LYMPHOCYTES, MCV, NM BKR ABSOLUTE NEUTROPHIL` — 1-2 negative values each (impossible for any of those measures)

A generic check was built to find these types of errors specifically.

# Medications: names fragmented, doses mostly clean

Only 17 distinct medication strings total — but `CellCept` (192), `mycophenolate` `mofetil` (211), and `MMF` (189) are the same drug under three names.

- **Dose:** only 24 distinct strings; a few look like intentionally injected errors — negative doses, `999 tablets daily`, `250 g twice daily` (likely a mg/g), `500 mL tablets daily` (a volume unit on a solid dosage form)
- **Two distinct, non-overlapping gaps:** 20 rows have dose but no medication name; 85 rows have medication name but no dose.
- Medication is normalized on every read (split to canonical medication name, value, unit, frequency)

# Design Decisions and Tradeoffs

# HIPAA controls, implemented literally

Real **login**, real **encryption at rest**, real **audit logging** — despite the data being confirmed synthetic/non-PHI and the deliverable requiring a public repo + public app link.



# Encrypt the store, not the source CSVs

- Keeps the repo **clone-and-run reproducible** — a reviewer can inspect the actual source data directly
- Still gives a real “encryption at rest” component to explain: the runtime SQLite store the app actually reads from is encrypted

# SQLite with a Subject supertype, not pandas end-to-end

```
subjects(subject_id PK, cohort)
├─ demographics(subject_id PK/FK)  ┌ ssc_patient only
├─ ssc_subtype (subject_id PK/FK)  └
└─ 9 clinical tables (subject_id FK) both cohorts
```

Models the **Control Subject / Registry** relationship correctly — cohort membership, not a foreign-key violation.

# Scope under a 2-day deadline

- Flat package layout over a layered architecture
- Minimal-but-real CI: ruff + mypy + pytest via GitHub Actions
- Targeted tests over exhaustive coverage

# Future Work

Known — not bugs. The app is stable, tested, deployed, green.

- Event overlay (BAL/skin-biopsy dates) on trajectory charts
- Medications are not classified (disease-modifying vs. symptomatic)
- Convert raw values to categorical variables with thresholds
- Make more flexible data visualization (boxplot vs violin)
- Add direct table views.

# Thank you — discussion

Repo: [github.com/sleyn/nw\\_ssc\\_data\\_engineering\\_prototype](https://github.com/sleyn/nw_ssc_data_engineering_prototype)

App: [nw-ssc-de.streamlit.app](https://nw-ssc-de.streamlit.app)

# Appendix

# Appendix: HIPAA-style controls, in detail

- **Login:** gated by a published demo account ([demo](#) / [SscDemo2026!](#)) — a demonstration of the control, not a real access boundary
- **Encryption at rest:** the runtime SQLite store is built encrypted from the source CSVs at startup; source CSVs themselves are left plaintext for reviewer reproducibility
- **Audit logging:** every login attempt (success/failure) appended to a local [audit.log](#) with timestamp and user; does not persist across restarts under the chosen ephemeral hosting (documented limitation)
- Full reasoning: [docs/adr/0001-hipaa-controls-with-public-demo-credentials.md](#), [docs/adr/0002-\\*.md](#), [docs/adr/0004-\\*.md](#)